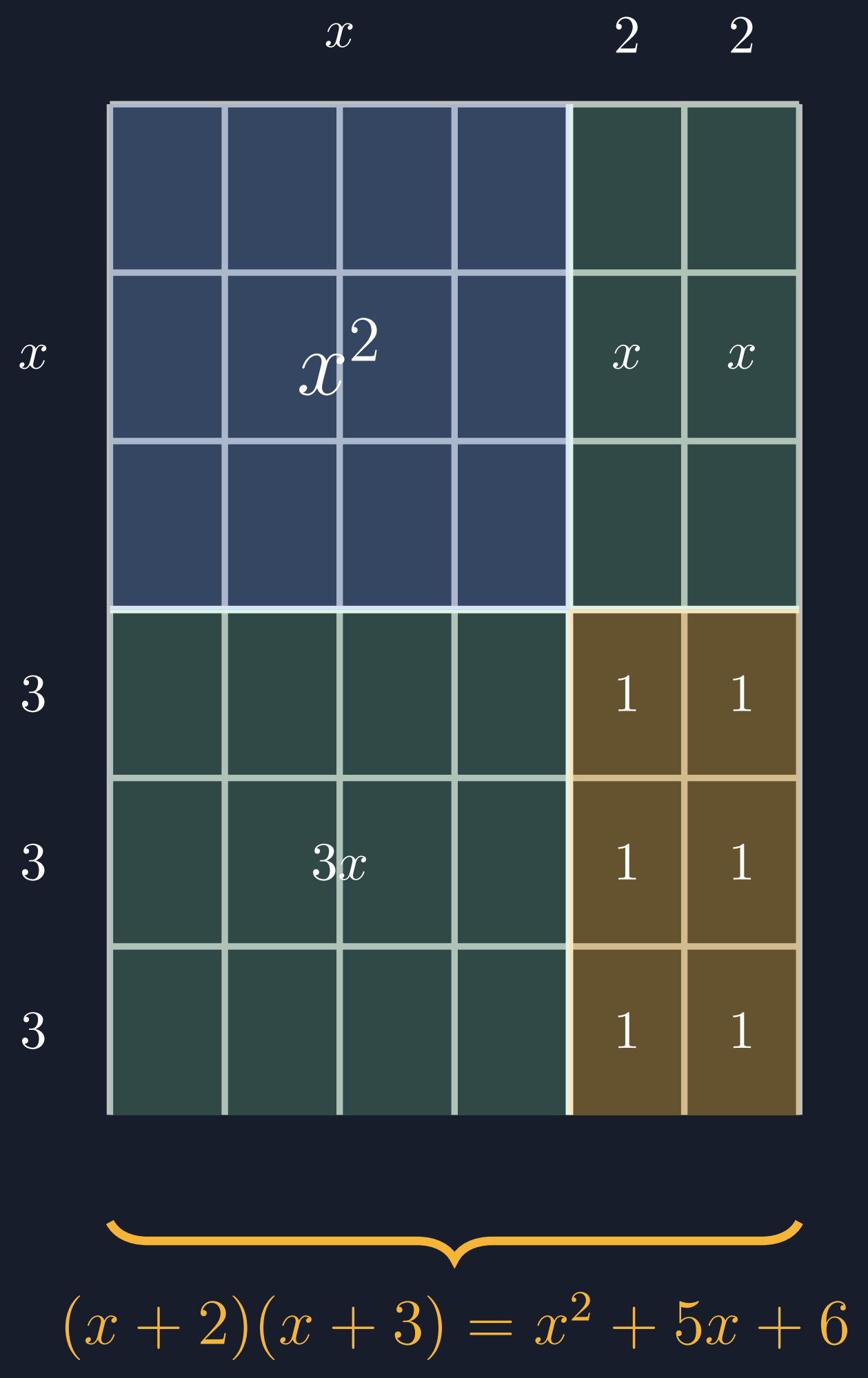

$$(x+2)(x+3) = x^2 + 5x + 6$$